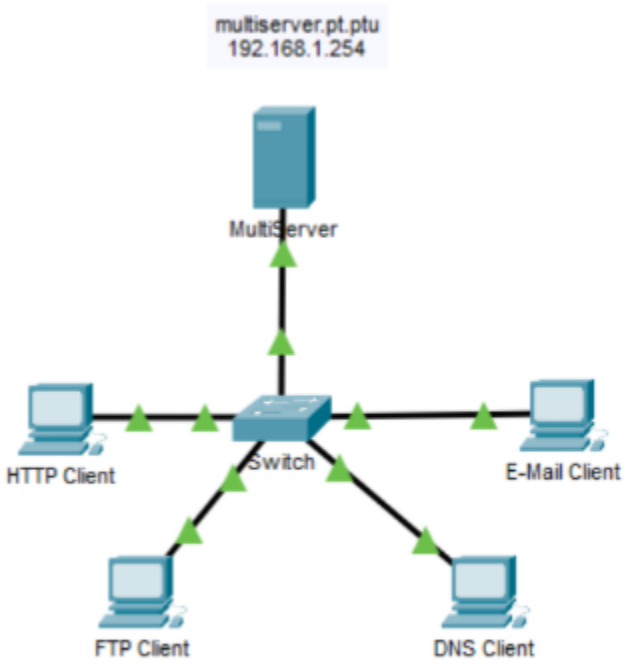
สิรภัทร ลีล้าน 673380067-2 sec2

Packet Tracer – TCP and UDP Communications

Objectives

Part 1: Generate Network Traffic in Simulation Mode
Part 2: Examine the Functionality of the TCP and UDP Protocols



Part 1: Generate Network Traffic in Simulation Mode and View Multiplexing

Step 1: Generate traffic to populate ARP tables

Perform the following task to reduce the amount of network traffic viewed in the simulation.

a. Click MultiServer and click the Desktop tab > Command Prompt
b. Enter the command:

```
ping -n 1 192.168.1.255
```

You are pinging the broadcast address for the client LAN.

The command option will send only one ping request rather than the usual four. This will take a few seconds as every device on the network responds to the ping request from MultiServer.

c. Close the MultiServer window.

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping -n 1 192.168.1.255

Pinging 192.168.1.255 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.255:
    Packets: Sent = 1, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Step 2: Generate Web (HTTP) traffic

a. Switch to Simulation mode
b. Click HTTP Client and open the Web Browser from the Desktop
c. In the URL field, enter:

```
192.168.1.254
```

and click Go

PDUs will appear in the topology window.

d. Minimize, but do not close, the HTTP Client configuration window.

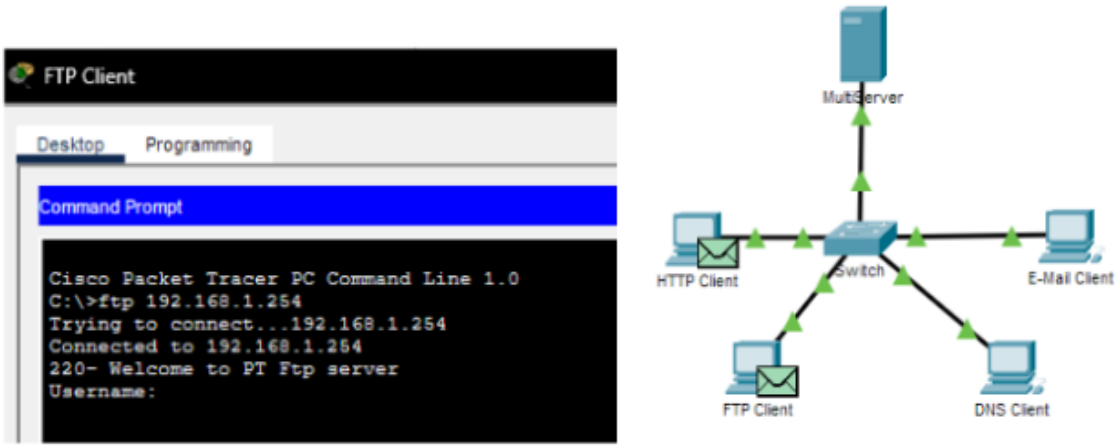
Step 3: Generate FTP traffic

a. Click FTP Client and open Command Prompt from Desktop
b. Enter the command:

```
ftp 192.168.1.254
```

PDUs will appear in the simulation window.

c. Minimize, but do not close, the FTP Client configuration window.



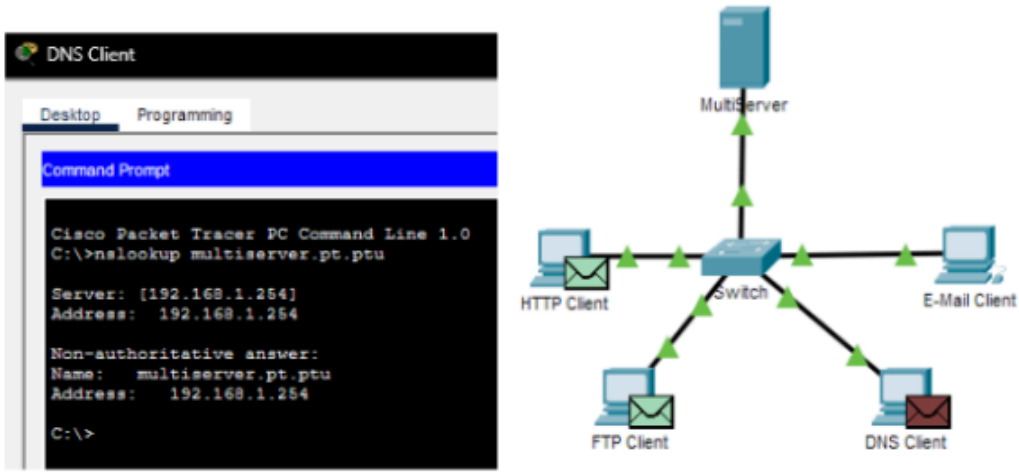
Step 4: Generate DNS traffic

a. Click DNS Client and open Command Prompt
b. Enter the command:

```
nslookup multiserver.pt.ptu
```

A PDU will appear in the simulation window.

c. Minimize, but do not close, the DNS Client configuration window.



Step 5: Generate Email traffic

a. Click E-Mail Client and open the E-Mail tool from Desktop
b. Click Compose and enter the following information:

To:

`user@multiserver.pt.ptu`

Subject:

`personalize the subject line`

Body:

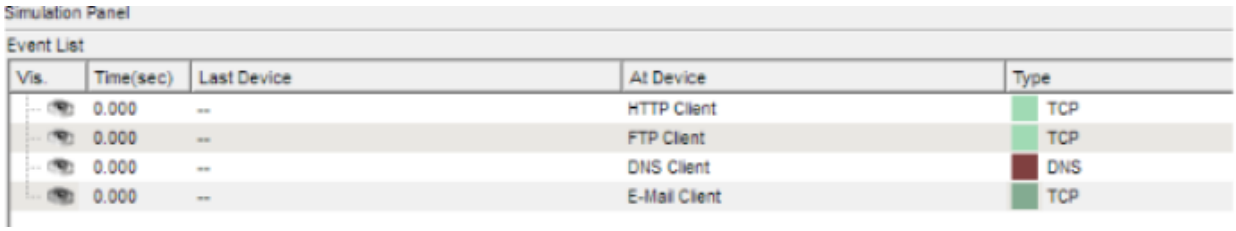
`personalize the Email`

c. Click Send
d. Minimize, but do not close, the E-Mail Client configuration window.

Step 6: Verify traffic is ready

There should now be PDU entries in the simulation panel for each client computer:

- HTTP Client – TCP
- FTP Client – TCP
- DNS Client – DNS (UDP)
- E-Mail Client – TCP



Step 7: Examine multiplexing

Observe how different protocol PDUs travel through the switch and server at the same time.

Multiplexing allows multiple applications to use the network simultaneously.

Part 2: Examine Functionality of TCP and UDP Protocols

Step 1: Examine HTTP traffic

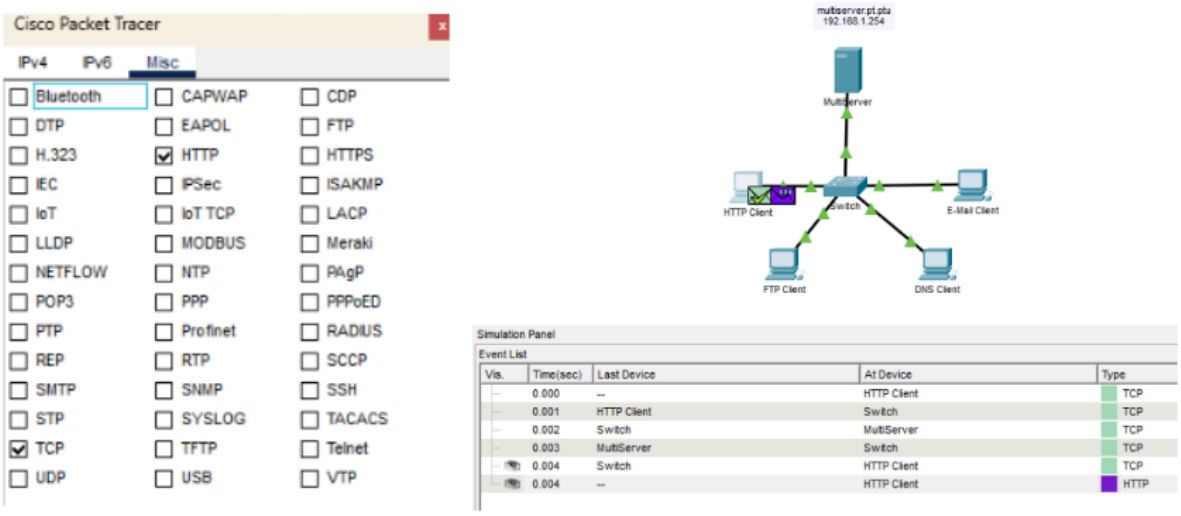
Why did it take so long for the HTTP PDU to appear?

Because TCP must complete the TCP 3-way handshake before sending HTTP data.

TCP handshake steps:

1. SYN
2. SYN-ACK
3. ACK

After handshake completes, HTTP data is transmitted.



What is the section labeled?

TCP

Because HTTP (Layer 7) runs on TCP (Layer 4).

Are these communications reliable?

Yes.

TCP is reliable because:

- Uses acknowledgements
- Retransmits lost packets
- Ensures correct order

Unlike UDP, which does not guarantee delivery.

Record values

SRC PORT: 1028
DEST PORT: 80
SEQUENCE NUM: 1
ACK NUM: 1

Which TCP flags are set?

ACK, PSH

How are port and sequence numbers different than before?

Ports are reversed:

Before:
Client → Server
Source Port: 1025
Destination Port: 80

After:
Server → Client
Source Port: 80
Destination Port: 1025

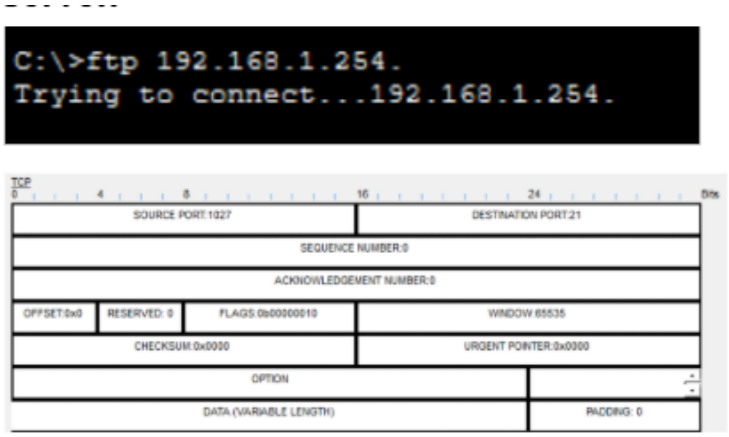
Sequence numbers change according to communication direction.

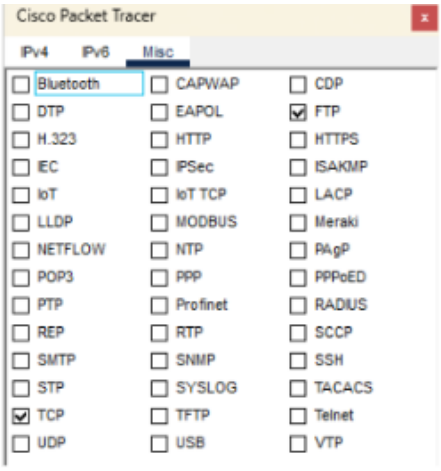
TCP section information

Source Port: 1025
Destination Port: 80
Sequence Number: 1
Acknowledgement Number: 1
Flags: PSH, ACK

Step 2: Examine FTP traffic

FTP also uses TCP.





Are communications reliable?

Yes.

FTP uses TCP which provides reliable communication.

Record values

SRC PORT: 1026
DEST PORT: 21
SEQUENCE NUM: 1
ACK NUM: 53
FLAG: FIN, ACK

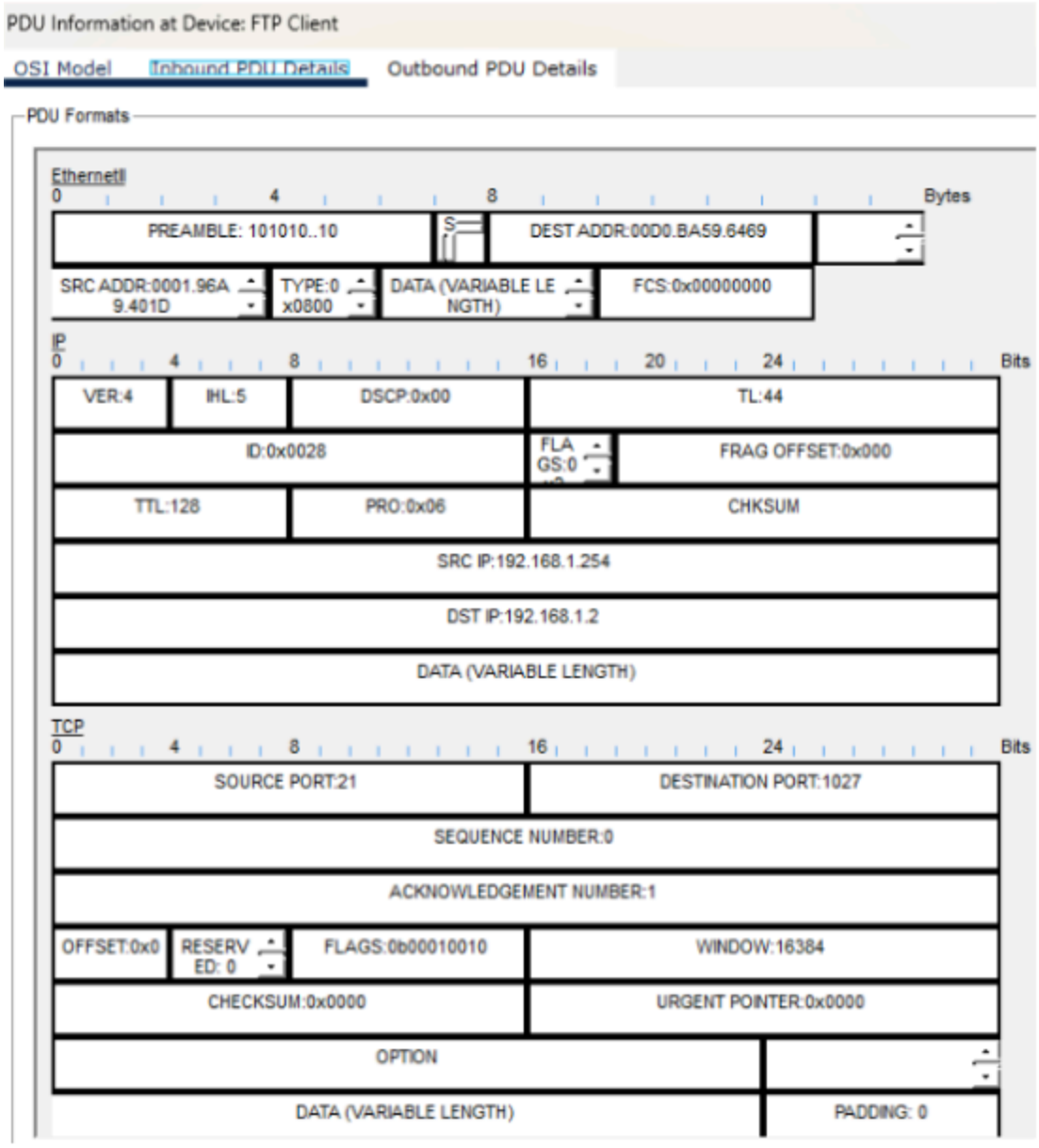
How are port and sequence numbers different than before?

Ports swap between client and server.

Client → Server:
Source Port: 1027
Destination Port: 21

Server → Client:
Source Port: 21
Destination Port: 1027

Sequence numbers update accordingly.



What is the message from server?

Welcome to the PT FTP server

Step 3: Examine DNS traffic

What is the Layer 4 protocol?

UDP

Are communications reliable?

No.

UDP is connectionless and unreliable because:

- No acknowledgement
- No retransmission
- No handshake

Record values

SRC PORT: 1033
DEST PORT: 53

Why no sequence and acknowledgement numbers?

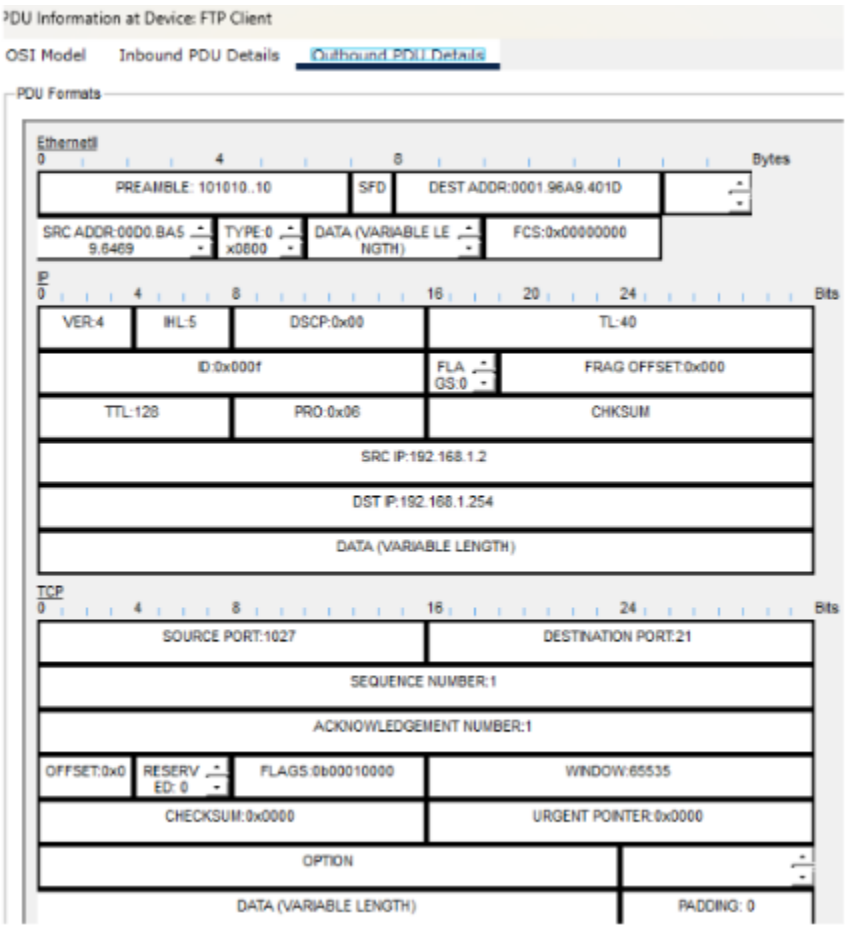
Because UDP does not use sequence numbers or acknowledgements.

Only TCP uses them.

How are port numbers different than before?

Client → Server:
Source Port: 1033
Destination Port: 53

Server → Client:
Source Port: 53
Destination Port: 1033



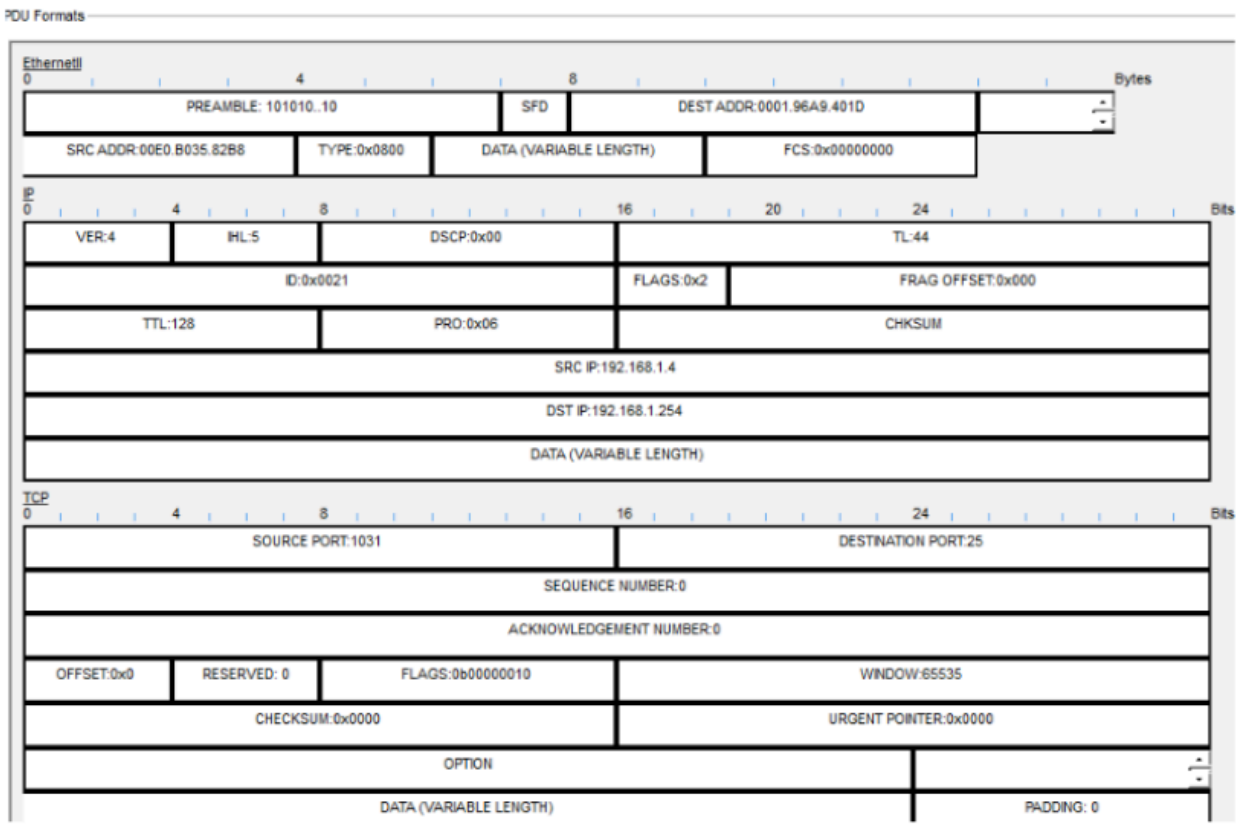
What is the last section of PDU called?

DNS Answer

What is the IP address for multiserver.pt.ptu?

192.168.1.254

Step 4: Examine Email traffic



What transport layer protocol is used?

TCP

Are communications reliable?

Yes.

TCP ensures reliable delivery.

Record values

SRC PORT: 1031
DEST PORT: 25
SEQUENCE NUM: 0
ACK NUM: 0
FLAG: SYN

How are ports and sequence numbers different?

Ports swap between client and server.

Client → Server
Source Port: 1031
Destination Port: 25

Server → Client
Source Port: 25
Destination Port: 1031

Sequence and acknowledgement numbers increase after handshake.

What email protocol is associated with TCP port 25?

SMTP (Simple Mail Transfer Protocol)

Used for sending email.

What email protocol is associated with TCP port 110?

POP3 (Post Office Protocol v3)

Used for receiving email.